

LAB WEEK 10

CORRELATION AND REGRESSION

DAILY DIARY STUDY OF MOOD

This study delivered a daily diary to 61 HIV infected men who have sex with men (MSM) between 16 and 24 years old for 66 days to measure HIV-risk behaviors and other psychosocial variables. The study examined the association between daily life stressors and daily negative mood. The study also examined the person-level association between the average number of daily life stressors across the 66 days and the average level of negative mood across that period.



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Ramirez, M, Wilson, P., Mitchell, R., Enders, C., & Woller, M. (in progress). Daily variability in depressed mood among gay and bisexual youth living with HIV. *Manuscript in preparation.*

KEY VARIABLES



Daily Life Stressors

Respondents were presented with list of stressful events (e.g., fights with family or friends, work stress, financial stress), and they checked how many they experienced each day.



Depression

Depression was measured using the Profile of Mood States (POMS). The POMS Depression-Dejection scale is computed by summing responses to five questionnaire items, where higher ratings reflect more negative mood.

RESEARCH QUESTION

- Question: Is there an association or trend between one's average number of life stressors and their average depressive mood?
- As the number of stressors increases or decreases, does depression change?
- Correlation and regression are appropriate for identifying trends between two numeric variables

LOAD PACKAGES AND IMPORT DATA

-  = data frame name
-  = variable name
-  = raw data file name

```
# LOAD R PACKAGES ----
```

```
# load R packages
library(GGally)
library(ggplot2)
library(psych)
library(summarytools)
```

```
# READ DATA ----
```

```
# github url for raw data
filepath <- 'https://raw.githubusercontent.com/craigenders/psych250a/main/data/DiaryData.csv'

# create data frame called ClinicalTrial from github data
Diary <- read.csv(filepath, stringsAsFactors = T)
```

SUMMARIZING DATA

 = data frame name
 = variable name

```
# INSPECT DATA ----
```

```
# summarize entire data frame (summarytools package)  
dfSummary()
```


R OUTPUT

Data Frame Summary
Diary
Dimensions: 61 x 2
Duplicates: 0

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
1	NumStressors [numeric]	Mean (sd) : 1.2 (1.1) min < med < max: 0 < 0.9 < 4.5 IQR (CV) : 1.6 (0.9)	60 distinct values	: : : : . : : : . . : : : : : : . .	61 (100.0%)	0 (0.0%)
2	Depress [numeric]	Mean (sd) : 6.2 (2.1) min < med < max: 4 < 5.6 < 12.9 IQR (CV) : 2.6 (0.3)	61 distinct values	: : : . : : . : : : : : . :	61 (100.0%)	0 (0.0%)

SCATTERPLOTS AND CORRELATIONS

 = data frame name
 = variable name

```
# SUBSET DATA: SELECT VARIABLES TO CORRELATE ----
```

```
# select variables to correlate
```

```
vars2corr <- Diary[, c('NumStressors', 'Support', 'Depress', 'Fatigue', 'WellBeing')]
```

```
# SCATTERPLOT MATRIX ----
```

```
# create the scatterplot matrix (GGalley package)
```

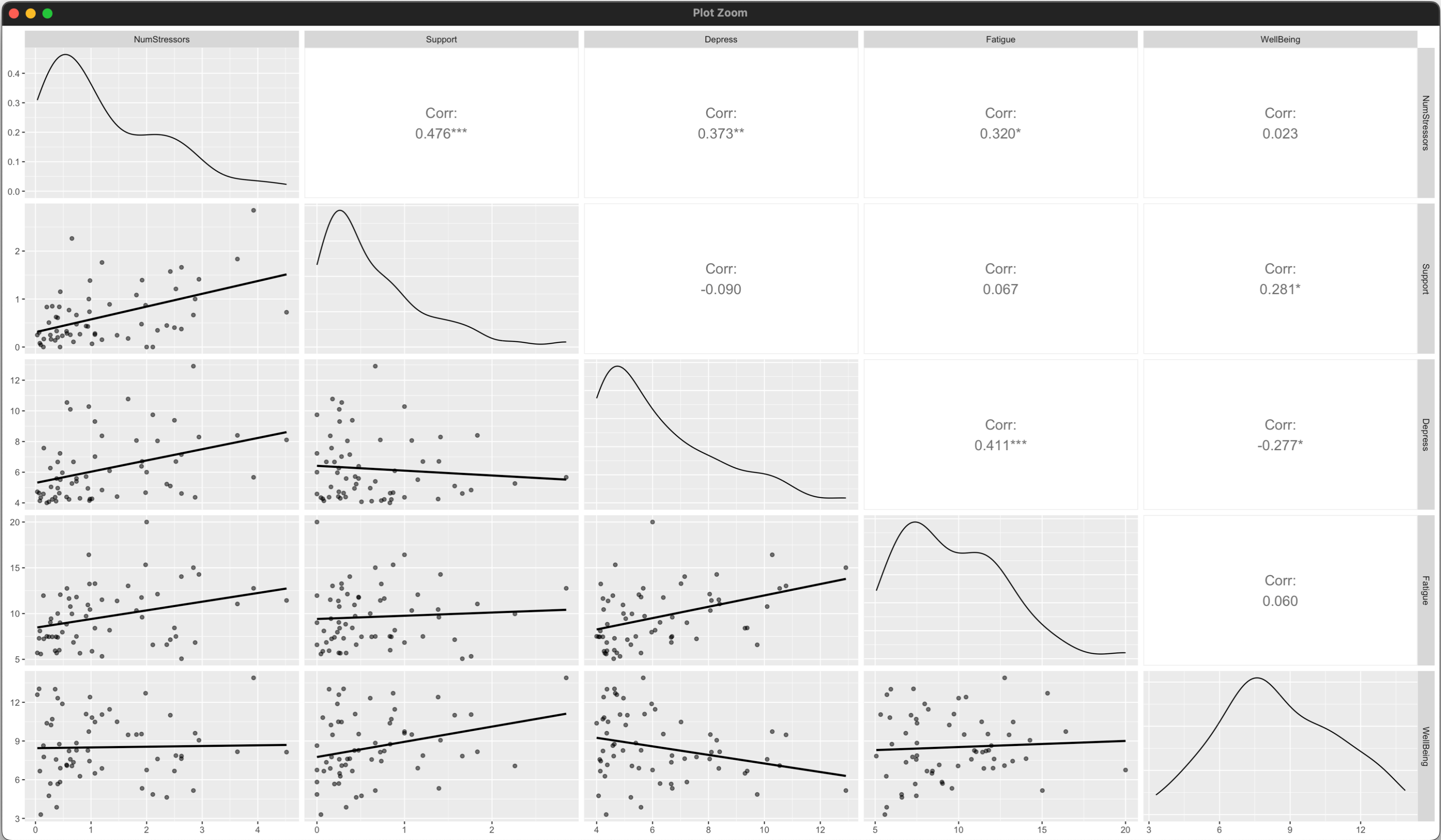
```
ggpairs(vars2corr,  
        upper = list(continuous = wrap('cor', size = 5)),  
        lower = list(continuous = wrap('smooth', method = 'lm', se = F, alpha = 0.6)),  
        diag = list(continuous = wrap('densityDiag')))
```

```
# CORRELATION MATRIX ----
```

```
# correlation matrix, confidence intervals, and p-values (psych package)
```

```
print(corr.test(vars2corr), short = FALSE)
```


R OUTPUT



R OUTPUT

Correlation matrix					
	NumStressors	Support	Depress	Fatigue	WellBeing
NumStressors	1.00	0.48	0.37	0.32	0.02
Support	0.48	1.00	-0.09	0.07	0.28
Depress	0.37	-0.09	1.00	0.41	-0.28
Fatigue	0.32	0.07	0.41	1.00	0.06
WellBeing	0.02	0.28	-0.28	0.06	1.00

Sample Size
[1] 61

Probability values (Entries above the diagonal are adjusted for multiple tests.)					
	NumStressors	Support	Depress	Fatigue	WellBeing
NumStressors	0.00	0.00	0.02	0.08	1.00
Support	0.00	0.00	1.00	1.00	0.17
Depress	0.00	0.49	0.00	0.01	0.17
Fatigue	0.01	0.61	0.00	0.00	1.00
WellBeing	0.86	0.03	0.03	0.65	0.00

R OUTPUT

Confidence intervals based upon normal theory. To get bootstrapped values, try `cor.ci`

	raw.lower	raw.r	raw.upper	raw.p	lower.adj	upper.adj
NmStr-Spprt	0.25	0.48	0.65	0.00	0.15	0.71
NmStr-Dprss	0.13	0.37	0.57	0.00	0.03	0.64
NmStr-Fatig	0.07	0.32	0.53	0.01	-0.02	0.59
NmStr-WllBn	-0.23	0.02	0.27	0.86	-0.23	0.27
Spprt-Dprss	-0.33	-0.09	0.17	0.49	-0.40	0.23
Spprt-Fatig	-0.19	0.07	0.31	0.61	-0.24	0.36
Spprt-WllBn	0.03	0.28	0.50	0.03	-0.06	0.56
Dprss-Fatig	0.18	0.41	0.60	0.00	0.07	0.66
Dprss-WllBn	-0.49	-0.28	-0.03	0.03	-0.55	0.05
Fatig-WllBn	-0.20	0.06	0.31	0.65	-0.23	0.34

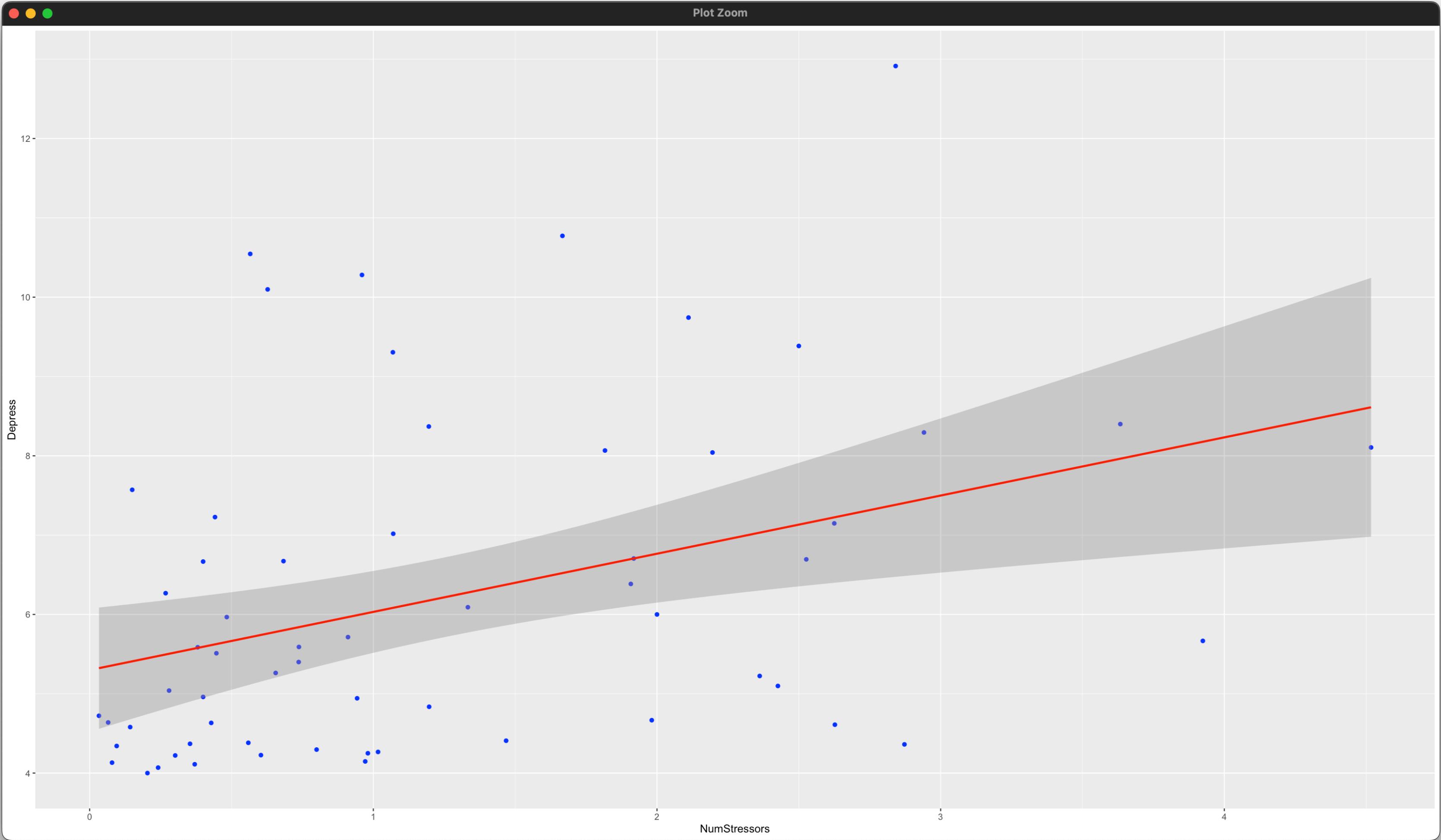
SCATTERPLOT AND REGRESSION LINE

-  = data frame name
-  = variable name
-  = predictor variable

```
# SCATTERPLOT WITH REGRESSION LINE ----
```

```
# scatterplot with regression line (ggplot2 package)
ggplot(data = Diary, aes(x = NumStressors, y = Depress)) +
  geom_point(color = 'blue') +
  geom_smooth(method = 'lm', color = 'red')
```

R OUTPUT



LINEAR REGRESSION

-  = data frame name
-  = variable name
-  = predictor variable

```
# LINEAR REGRESSION WITH 95% CONFIDENCE INTERVALS ----
```

```
# linear regression (base R)
```

```
results <- lm(Depress ~ NumStressors, data = Diary)
```

```
summary(results)
```

```
# 95% confidence intervals (base R)
```

```
confint(results)
```


R OUTPUT

Residuals:

Min	1Q	Median	3Q	Max
-3.0436	-1.3404	-0.4641	0.8722	5.5312

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	5.2988	0.3877	13.666	< 2e-16	***
NumStressors	0.7334	0.2371	3.093	0.00303	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Intercept and slope

Residual standard error: 1.96 on 59 degrees of freedom

Multiple R-squared: 0.1395, Adjusted R-squared: 0.1249

F-statistic: 9.565 on 1 and 59 DF, p-value: 0.003028

F and R-square

	2.5 %	97.5 %
(Intercept)	4.5229278	6.074675
NumStressors	0.2588819	1.207889

95% confidence intervals



SMALL GROUP EXERCISE

Download two files from Bruin Learn: “Week 10 Lab. Correlation and Regression.R” and “Week 10 Small Group Exercise.R”. The Lab script contains the R code we just discussed. The Exercise script contains only the URL for a different data set, Discrimination.csv. In groups of two or three, you will complete a series of R tasks that provide practice for the next assignment. There is no need to write code from scratch; instead, you can copy and paste code chunks from the Lab file into your Exercise script, modifying the data and variable names as needed. The Discrimination.csv file for this exercise contains data from a study investigating the association between perceived discrimination and internalizing behaviors.

DISCRIMINATION AND INTERNALIZING BEHAVIOR

There is a well-documented relationship between perceived discrimination and internalizing symptoms among Latinx adolescents. However, few studies have examined how this psychosocial stressor relates to multiple domains of functioning in rural Latinx adolescents simultaneously. This study tested a spillover model of internalizing symptom development, where the negative effects of perceived discrimination are experienced through peer and family relationships.



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Ramos, G., Delgadillo, D., Fossum, J., Montoya, A.K., Thamrin, H., Rapp, A., Escovar, E., Chavira, D.A. (2021). Discrimination and internalizing symptoms in rural Latinx adolescents: An ecological model of etiology. *Children and Youth Services Review*, 130, 1-10.

KEY VARIABLES



Perceived Discrimination

Perceived discrimination refers to individuals' perception of negative attitude, judgment, or unfair treatment due to their specific characteristics such as gender, race, ethnicity, and social status.



Internalizing Behavior

Internalizing behaviors are behaviors directed inwardly toward oneself. Examples include anxious and depressive symptoms, social withdrawal, and somatic (physical) complaints.



SMALL GROUP EXERCISE TASK 1

- Use the provided URL to import the `Discrimination.csv` file into an R data frame (import method #3 from the Week 0 lab script).
- Use the `dfSummary` function to get numeric and visual summaries of the data frame's variables.



SMALL GROUP EXERCISE TASK 2

- Use the `ggpairs` function in the `Ggally` package to obtain scatterplots among the following variables: Age, Discrimination, and Internalizing.
- Use the `corr.test` function from the `psych` package to obtain a correlation matrix.
- Consider the correlation between Discrimination and Internalizing. Provide an interpretation of the correlation and discuss its direction and strength.



SMALL GROUP EXERCISE TASK 3

- Use the ggplot2 package to obtain a scatterplot and regression line from an analysis where Discrimination is the predictor and Internalizing is the outcome.
- Use the lm function to perform a regression analysis where Discrimination is the predictor and Internalizing is the outcome.
- Provide an interpretation of the intercept and slope coefficients from the regression analysis.



SMALL GROUP EXERCISE TASK 4

- Decide on the null hypothesis that perceived discrimination and internalizing symptoms are unrelated. Based on the p-value, state whether you reject or fail to reject H_0 . Justify your decision with both statistical reasoning and a short statement about the practical meaning.
- In addition to determining whether perceived discrimination has a statistically significant effect on internalizing symptoms, evaluate the strength of the relationship by interpreting the R^2 effect size. Explain what the numeric value means and compare it to Cohen's commonly used benchmarks.